Review

Scenario 1: σ is known.

① $X \sim N(\mu, \sigma^2)$

② $X \overset{\cdot}{\sim} (\mu, \sigma^2)$ $n \geq 30$

μ is unknown.

And $(1-\alpha)100\%$ CI for μ.

$$\bar{x} \pm Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}.$$

Scenario 2: σ is unknown.

①. $X \sim N(\underbrace{\mu, \sigma^2}_{\text{unknown}})$

② $X \sim \cdot (\mu, \sigma^2)$ $n \geq 30$

$(1-\alpha)100\%$ CI for μ.

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}.$$

Example: Air conditioning ~~cooling~~ coil temperature follow a normal distribution $N(\mu, \sigma^2)$

a) suppose 15 units are randomly selected.

$\bar{x} = 3.4$ $\quad S = 0.9$

Find 99% CI for μ. $\quad (1-\alpha) = 0.99 \Rightarrow \alpha = 0.01$

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} = 3.4 \pm \underbrace{t_{0.005, 14}} \frac{0.9}{\sqrt{15}} = 3.4 \pm 2.977 \frac{0.9}{\sqrt{15}}$$

$$= (2.708, 4.092)$$

b) Supposed 1000 units selected

$\bar{x} = 3.5$ $\quad S = 1.0$

Find a 95% CI for μ. $\quad \alpha = 0.05$

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} = 3.5 \pm \underbrace{t_{0.025, 999}} \frac{1}{\sqrt{1000}} = 3.5 \pm 1.96 \frac{1}{\sqrt{1000}}$$

$$= (3.438, 3.562)$$

If df > 100, we can approximate $t_{\frac{\alpha}{2}, n-1}$ with $Z_{\frac{\alpha}{2}}$.